Layer 1: Stochastic Demand Forecasting

Document Owner: Zernitsky Itamar **Status:** Design Reference **Scope:** Complete technical specification of all six stages of the stochastic demand forecasting pipeline: phase-aligned history synthesis, attention pooling, feature engineering, XGBoost point forecasts, hierarchical Bayesian variance model, scenario generation, and calibration.

> **What this layer does.** The existing engine scales historical demand by α and produces one number per provider. This layer replaces that single number with 200 plausible demand futures — scenarios — drawn from a calibrated posterior predictive distribution. The 200 scenarios flow directly into the Layer 2 SAA chance constraints, replacing the vague "minimize deviation" objective with a formal probabilistic coverage guarantee.

> **Reading note.** Every section follows the same consistent SITE-001 example dataset: 13 historical weeks, 4 OR rooms, 18 providers (Garcia, Chen, Thompson, Wright, Kim, Torres, Patel, Jones, and 10 others). The same anomalies — Garcia's Thursday exception in W01, Wright's Friday cover in W05 — appear throughout so every formula connects back to the same ground truth.

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Stage 0: Phase-Aligned History Synthesis

What It Does

Builds the weekly observation matrix from raw EHR case data, aligned to the rotation phase detected by the Pre-Layer. This is the foundational data transformation — everything downstream depends on getting this right.

The key insight: OR demand is not a stationary time series. A provider on a 2-week rotation behaves differently in odd weeks (Phase 0) than in even weeks (Phase 1). If we compute a naive 4-week trailing mean, we average across two structurally different conditions. Phase-aligned windows fix this by looking back only at same-phase weeks.

Inputs

Field	Source	Description
block_case_minutes	EHR per case	Surgery minutes inside assigned block
turnover_time	EHR per case	Room reset time between consecutive cases
open_block bool	EHR per case	Whether case occurred outside provider's block
manual_early_release	EHR per block	Minutes released back to OR early
T* and phase_labels	Pre-Layer	Rotation period and phase tag per week
providers_in_opt	Input params	Which providers need scenario generation

Mechanism

For each provider \$p\$ and day of week d, aggregate EHR records into weekly observations:

In-block case minutes per week:

$$\text{CaseTime}_{p,d,w} = \sum_{c \in \text{cases}(p,d,w)} \text{block\_case\_minutes}_c$$

Turnover minutes per week:

$$\text{Turnover}_{p,d,w} = \sum_{c \in \text{cases}(p,d,w)} \text{turnover\_time}_c$$

Utilization per week (with early release in denominator):

$$\text{util}_{p,d,w} = \frac{\text{CaseTime}_{p,d,w} + \text{Turnover}_{p,d,w}}{\text{AllocatedBlockTime}_{p,d,w} - \text{EarlyRelease}_{p,d,w}}$$

EarlyRelease projected forward (deterministic — no scenario subscript):

$$\text{EarlyRelease}_{\text{projected}}[p] = \alpha \cdot \sum_{b \in B_p^{\text{hist}}} \text{EarlyReleaseDur}_b$$

where $\alpha = T_{\text{opt}} / T_{\text{hist}}$ is the window scaling ratio.

Each weekly observation is then tagged with its rotation phase $\phi = (w + k^*) \bmod T^*$ from the Pre-Layer.

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║  PHASE-ALIGNED WEEKLY OBSERVATIONS  ·  SITE-001  ·  Garcia × Tuesday          ║
╠═══════════════════════════════════════════════════════════════════════════════╣
║                                                                               ║
║  RAW AGGREGATION (before phase alignment)                                     ║
║  ──────────────────────────────────────────────────────────────────────────   ║
║  Week │ Phase │ CaseTime │ Turnover │ EarlyRel │ AllocBlock │ util_pct        ║
║  ─────┼───────┼──────────┼──────────┼──────────┼────────────┼──────────       ║
║  W00  │   0   │   84 min │  14 min  │   28 min │  480 min   │  21.7%          ║
║  W01  │   1   │   78 min │  13 min  │   22 min │  480 min   │  19.8%          ║
║  W02  │   0   │   73 min │  12 min  │   30 min │  480 min   │  18.6%          ║
║  W03  │   1   │   69 min │  11 min  │   25 min │  480 min   │  17.5%          ║
║  W04  │   0   │   82 min │  13 min  │   29 min │  480 min   │  19.8%          ║
║  W05  │   1   │   65 min │  10 min  │   20 min │  480 min   │  16.5%          ║
║         ↑ Wright covered this week — Garcia had 0 case mins                   ║
║  W06  │   0   │   61 min │  10 min  │   32 min │  480 min   │  15.6%          ║
║  W07  │   1   │   74 min │  12 min  │   24 min │  480 min   │  18.8%          ║
║  W08  │   0   │   69 min │  11 min  │   28 min │  480 min   │  17.5%          ║
║  W09  │   1   │   65 min │  10 min  │   21 min │  480 min   │  16.5%          ║
║  W10  │   0   │   68 min │  11 min  │   30 min │  480 min   │  17.3%          ║
║  W11  │   1   │   69 min │  11 min  │   27 min │  480 min   │  17.5%          ║
║  W12  │   0   │   68 min │  11 min  │   20 min │  480 min   │  17.3%          ║
║                                                                               ║
║  PHASE-SEPARATED VIEWS                                                        ║
║  ──────────────────────────────────────────────────────────────────────────   ║
║  Phase 0 weeks (W00,W02,W04,W06,W08,W10,W12):                                 ║
║    CaseTime mean: 72.1 min/wk    util mean: 18.2%                             ║
║                                                                               ║
║  Phase 1 weeks (W01,W03,W05,W07,W09,W11):                                     ║
║    CaseTime mean: 70.0 min/wk    util mean: 17.8%                             ║
║    (slightly lower — W05 Wright cover drags the mean down)                    ║
║                                                                               ║
║  KEY INSIGHT: Naive 4-week trailing mean for W12 =                            ║
║    avg(W09, W10, W11, W12) = avg(16.5, 17.3, 17.5, 17.3) = 17.2%              ║
```

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║  Phase-aligned 4-week trailing mean for W12 (phase 0) =                   ║
║    avg(W04, W06, W08, W10) = avg(19.8, 15.6, 17.5, 17.3) = 17.6%          ║
║  Small difference here because Garcia is stable.                          ║
║  For providers with strong phase effects (Thompson/Patel) the difference  ║
║  is the difference between a meaningful feature and pure noise.           ║
║                                                                           ║
║  EARLY RELEASE PROJECTION  (deterministic)                                ║
║  ─────────────────────────────────────────────                            ║
║  Garcia hist. total early release: 28+22+30+25+29+20+32+24+28+21+30+27    ║
║                                  = 316 min over 13 weeks                  ║
║  α = T_opt / T_hist = 13/13 = 1.00  (equal window lengths)                ║
║  EarlyRelease_projected[Garcia] = 1.00 × 316 = 316 min = 5.27 hrs         ║
║                                                                           ║
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```

Outputs

- `df_weekly` — DataFrame: provider_id, service_line, week_start, rotation_phase, day_of_week, casetime_min, turnover_min, early_release_min, out_of_block_min, util_pct, is_included
- `early_release_projected` — per provider: deterministic scalar in minutes

Stage 1: Attention Pooling

Why Attention Over a Simple Trailing Mean

A naive 4-week trailing mean of util_pct treats all 4 weeks equally. But not all weeks carry the same information about next week's demand. A week where Garcia was covered by Wright (W05) is an exception, not a signal about Garcia's own demand. A week from 10 weeks ago is less informative about today's Garcia than last week's Garcia.

The attention pooler learns **which weeks to weight more** when summarizing the 12-week history. It does this in a self-supervised way: train to predict next week's demand from a weighted summary of the past 12 weeks. The weights that minimize next-step prediction error are exactly the weights that carry the most forward-looking information.

Architecture

Single-query attention over a sequence of $T = 12$ phase-aligned weekly util_pct observations. The query is a single learnable scalar w_q shared across all providers — one number captures the global tendency to upweight or downweight recent/high-utilization weeks.

Score computation:

$$\text{score}_t = w_q \cdot x_t \quad \forall\, t \in 1, \ldots, 12$$

Attention weights (softmax normalization):

$$\alpha_t = \frac{\exp(\text{score}_t)}{\displaystyle\sum_{t'=1}^{12} \exp(\text{score}_{t'})}$$

Context vector (attended summary):

$$\text{context} = \textstyle\sum_{t=1}^{12} \alpha_t \cdot x_t$$

This is a degenerate but powerful form of self-attention: $K = V = x$ (the raw util_pct sequence serves as both keys and values) and $Q = w_q$ (a single scalar query). No projection matrices, no multi-head complexity. The inductive bias is correct for this problem: we want a weighted average of historical utilization where the weights are learned to maximize predictive value.

Training Procedure

Self-supervised objective: predict the next week's utilization from the attended history.

$$\mathcal{L} = \frac{1}{N_{\text{train}}} \textstyle\sum_{(p,d,t) \in \text{train}} \left(\sum_{\tau=1}^{12} \alpha_\tau \cdot x_{p,d,t-12+\tau} - x_{p,d,t} \right)^2$$

This is a next-step prediction task — the only label needed is the actual utilization in the next week, which is already in the training data. No manual labeling required.

Optimizer: numpy SGD with learning rate 0.01, 500 epochs, gradient clipping at 1.0. PyTorch optionally available but disabled by default to avoid JAX/SIGSEGV conflict on macOS.

Shared weights: w_q is shared across all providers and all days. This regularizes the model — we do not learn a different w_q per provider, which would overfit on sparse providers. The shared w_q learns the globally best attention pattern for OR demand.

Result on SITE-001: $w_q = 0.046$. Since $w_q > 0$, higher-utilization weeks receive higher scores and thus higher attention weights. The model has learned that high-utilization weeks are more predictive of the next week than low-utilization weeks — consistent with mean-reversion behavior in OR scheduling.

Three Output Features

attn_weighted_util — the characteristic operating level:

$$\text{attn\_weighted\_util} = \sum_{t=1}^{12} \alpha_t \cdot x_t$$

A better point summary than the simple mean. When $w_q > 0$ and a provider is trending upward (recent weeks higher than old weeks), the attention-weighted util is higher than the simple mean — the model correctly anticipates the upward trend. For Garcia (flat history), it nearly equals the simple mean.

attn_entropy — the consistency of the pattern:

$$\text{attn\_entropy} = -\sum_{t=1}^{12} \alpha_t \cdot \log(\alpha_t)$$

Maximum possible entropy for 12 weeks is $\ln(12) = 2.485$. When entropy is near-maximum, all weeks are equally informative — the pattern is stable and uniform. When entropy is low, a few weeks dominate — the provider has a volatile or recently-changed pattern. XGBoost uses entropy as a signal for how confident it should be in its prediction.

attn_recency_bias — the trend direction:

$$\text{attn\_recency\_bias} = \sum_{t=1}^{12} \alpha_t \cdot t_{\text{norm}}$$

where $t_{\text{norm}} \in [0, 1]$ runs from 0 (oldest) to 1 (most recent). Value above 0.5 means model attends more to recent weeks — the provider is improving. Value below 0.5 means the model attends more to older weeks — declining. Value near 0.5 means flat pattern.

Runs **twice** — once for CaseTime util sequence, once for Turnover util sequence — producing 6 attention features total per provider×day.

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║  ATTENTION POOLER  ·  SITE-001  ·  Selected providers  ·  Tuesday slot      ║
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║                                                                             ║
║  TRAINED WEIGHT:  w_q = 0.046                                               ║
║                                                                             ║
║  DR. JAMES GARCIA — Tuesday  (low-volume, flat)                             ║
║  ─────────────────────────────────────────────────────────────────────────  ║
║  CaseTime util sequence x (W01–W12, phase-aligned):                         ║
║  [0.21, 0.19, 0.18, 0.17, 0.20, 0.16, 0.15, 0.18, 0.17, 0.16, 0.17, 0.17]   ║
║                                                                             ║
║  Scores = w_q × x:                                                          ║
║  [0.0097, 0.0087, 0.0083, 0.0078, 0.0092, 0.0074, 0.0069, 0.0083,           ║
║   0.0078, 0.0074, 0.0078, 0.0078]                                          ║
║                                                                             ║
║  α = softmax(scores):                                                       ║
║  [0.0846, 0.0840, 0.0838, 0.0835, 0.0843, 0.0833, 0.0830, 0.0838,           ║
║   0.0835, 0.0833, 0.0835, 0.0835]   ← nearly uniform (all similar util)     ║
║                                                                             ║
║  attn_weighted_util  = Σ αₜ·xₜ       = 0.176     (chronic ~17% util)          ║
║  attn_entropy        = −Σ αₜ·log(αₜ) = 2.484     (near-max 2.485 = flat)      ║
║  attn_recency_bias   = Σ αₜ·t_norm   = 0.500     (no trend — flat)            ║
║                                                                             ║
║  INTERPRETATION: Garcia's history is uniformly bad. All weeks look the      ║
║  same to the model. High entropy = consistent pattern. Recency=0.5 = flat.  ║
║                                                                             ║
║  ─────────────────────────────────────────────────────────────────────────  ║
║  DR. SARAH CHEN — Tuesday  (high-volume, stable)                            ║
║  ─────────────────────────────────────────────────────────────────────────  ║
║  CaseTime util sequence x:                                                  ║
║  [0.88, 0.91, 0.89, 0.92, 0.90, 0.93, 0.91, 0.90, 0.88, 0.91, 0.90, 0.90].  ║
║                                                                             ║
║  attn_weighted_util  = 0.903     (consistently near 90%)                    ║
║  attn_entropy        = 2.483     (near-max — uniformly strong pattern)      ║
║  attn_recency_bias   = 0.501     (essentially no trend)                     ║
║                                                                             ║
║  ─────────────────────────────────────────────────────────────────────────  ║
║  HYPOTHETICAL IMPROVING PROVIDER — Tuesday  (trend upward)                  ║
║  ─────────────────────────────────────────────────────────────────────────  ║
║  util sequence x:                                                           ║
║  [0.30, 0.32, 0.35, 0.37, 0.40, 0.43, 0.46, 0.49, 0.52, 0.55, 0.58, 0.61].  ║
║                                                                             ║
║  w_q=0.046 gives higher scores to high-util (recent) weeks:                 ║
║    α_early  ≈ 0.065  (lower weight on old low-util weeks)                   ║
║    α_recent ≈ 0.095  (higher weight on new high-util weeks)                 ║
║                                                                             ║
║  attn_weighted_util  = 0.491     (above simple mean of 0.454 — trend ↑)     ║
║  attn_entropy        = 2.43      (slightly below max — non-uniform weights) ║
║  attn_recency_bias   = 0.557     (> 0.5 → recent weeks dominate → uptrend)  ║
║                                                                             ║
```

```
║  INTERPRETATION: attn_recency_bias=0.557 signals to XGBoost: "this        ║
║  provider is improving — weight recent history more." The naive trailing   ║
║  mean would give equal weight to the old 0.30 weeks, underestimating       ║
║  the provider's current trajectory.                                        ║
║                                                                            ║
║  TURNOVER ATTENTION  (same w_q applied to turnover util sequence)           ║
║  ─────────────────────────────────────────────────────────────────────────  ║
║  Garcia Turnover sequence:                                                 ║
║  [0.030,0.028,0.025,0.022,0.030,0.020,0.018,0.025,0.022,0.020,0.022,0.022]. ║
║                                                                            ║
║  attn_turn_weighted  = 0.024     (turnover ~2% of block time)              ║
║  attn_turn_entropy   = 2.484     (uniform pattern)                         ║
║  attn_turn_recency   = 0.499     (flat trend)                              ║
║                                                                            ║
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Outputs

- `attention_pooler.json` — saved w_q value (persists across quarters)
- Per provider×day: 6 attention features (3 CaseTime + 3 Turnover)

Stage 2: Feature Engineering (27 Features)

Feature Matrix Design

The 27-feature matrix is the input to both XGBoost models (CaseTime and Turnover). Features are organized into four groups:

Group A: Phase-Aligned Trailing Windows (9 features)

These are the core demand signal features. All trailing windows are computed within-phase — looking back at same-phase weeks only.

Feature	Window	Component	Description
trailing_4w_mean_case	4 weeks	CaseTime	Phase-aligned 4-week mean util_pct
trailing_12w_mean_case	12 weeks	CaseTime	Phase-aligned 12-week mean util_pct

Feature	Window	Component	Description
trailing_4w_std_case	4 weeks	CaseTime	Phase-aligned 4-week std dev
trailing_12w_std_case	12 weeks	CaseTime	Phase-aligned 12-week std dev
trailing_4w_min_case	4 weeks	CaseTime	4-week minimum (floor signal)
trailing_4w_max_case	4 weeks	CaseTime	4-week maximum (ceiling signal)
trailing_8w_mean_case	8 weeks	CaseTime	Mid-range window (intermediate)
trailing_4w_mean_turn	4 weeks	Turnover	Phase-aligned 4-week turnover mean
trailing_12w_mean_turn	12 weeks	Turnover	Phase-aligned 12-week turnover mean

Group B: Seasonality Features (4 features)

Fourier basis encoding week-of-year cyclicality:

$$\mathrm{sin\_woy} = \sin\left(\frac{2\pi \cdot \mathrm{woy}}{52}\right), \quad \mathrm{cos\_woy} = \cos\left(\frac{2\pi \cdot \mathrm{woy}}{52}\right)$$

$$\mathrm{sin\_woy\_2} = \sin\left(\frac{4\pi \cdot \mathrm{woy}}{52}\right), \quad \mathrm{cos\_woy\_2} = \cos\left(\frac{4\pi \cdot \mathrm{woy}}{52}\right)$$

Two harmonics capture both annual and semi-annual seasonal patterns in OR scheduling (summer slowdown, holiday reduction).

Group C: Structural Features (5 features)

Feature	Description
dow_Mon through dow_Fri	Day-of-week one-hot (5 binary features)
service_line_wk_mean	Average util across all providers in the same service line this week. Cross-provider context signal.
weeks_observed	How many weeks of history exist for this provider×day. Tenure proxy — low values flag sparse providers to XGBoost.
rotation_phase	One-hot phase label (0 or 1 for T*=2). Captures systematic

Feature	Description
	phase-level differences in demand.
exception_rate_series	Fraction of historical weeks with exception assignment for this block series. From Pre-Layer.

Group D: Attention Features (6 features)

Feature	Component	Description
attn_weighted_util	CaseTime	Characteristic operating level
attn_entropy	CaseTime	Pattern consistency
attn_recency_bias	CaseTime	Trend direction
attn_turn_weighted	Turnover	Turnover operating level
attn_turn_entropy	Turnover	Turnover consistency
attn_turn_recency	Turnover	Turnover trend

```
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║  FEATURE MATRIX  ·  SITE-001  ·  Garcia × Tuesday  ·  Week W12               ║
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║                                                                              ║
║  GROUP A: Phase-aligned trailing windows                                     ║
║  ─────────────────────────────────────────────────                           ║
║  trailing_4w_mean_case   =  0.171   (W04,W06,W08,W10 phase-0 mean)           ║
║  trailing_12w_mean_case  =  0.178   (all phase-0 weeks mean)                 ║
║  trailing_4w_std_case    =  0.016   (moderate variability)                   ║
║  trailing_12w_std_case   =  0.019                                            ║
║  trailing_4w_min_case    =  0.156   (W06 was the low week)                   ║
║  trailing_4w_max_case    =  0.198   (W04 was high)                           ║
║  trailing_8w_mean_case   =  0.174   (W00,W02,W04,W06,W08,W10,W12 mean)       ║
║  trailing_4w_mean_turn   =  0.022   (turnover ~2% of block)                  ║
║  trailing_12w_mean_turn  =  0.024                                            ║
║                                                                              ║
║  GROUP B: Seasonality                                                        ║
║  ─────────────────────────────────────────────────                           ║
║  sin_woy     =  0.784   (week 34 of year — late summer)                      ║
║  cos_woy     =  0.621                                                        ║
║  sin_woy_2   =  0.971                                                        ║
║  cos_woy_2   = -0.241                                                        ║
║                                                                              ║
║  GROUP C: Structural                                                         ║
║  ─────────────────────────────────────────────────                           ║
║  dow_Tue     =  1   (Tuesday indicator)                                      ║
║  dow_others  =  0                                                            ║
```

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║  service_line_wk_mean = 0.410  (Cardiac avg this week across site)          ║
║  weeks_observed       = 12    (complete history — not sparse)               ║
║  rotation_phase       = 0    (Phase 0 week, W12 is even)                    ║
║  exception_rate_series = 0.00  (Garcia's Tuesday is never contested)        ║
║                                                                             ║
║  GROUP D: Attention                                                         ║
║  ────────────────────────────────────────────────                           ║
║  attn_weighted_util  = 0.176   (confirms chronic ~17% CaseTime util)        ║
║  attn_entropy        = 2.484   (uniform history — no outlier weeks)         ║
║  attn_recency_bias   = 0.500   (no trend — flat)                            ║
║  attn_turn_weighted  = 0.024   (turnover signal)                            ║
║  attn_turn_entropy   = 2.483                                                ║
║  attn_turn_recency   = 0.499                                                ║
║                                                                             ║
║  TARGET VARIABLES (what XGBoost predicts):                                  ║
║  casetime_min  = 69.0   (actual W12 CaseTime minutes)                       ║
║  turnover_min  = 11.1   (actual W12 Turnover minutes)                       ║
║                                                                             ║
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Outputs

- `df_features` — 27-column feature matrix with target columns: casetime_min, turnover_min
- `features.parquet` — persisted artifact

Stage 3: XGBoost Point Forecasts

Architecture Design

Two separate XGBoost regressors are trained: one predicting weekly CaseTime minutes, one predicting weekly Turnover minutes. Separate models because CaseTime and Turnover have different patterns — a high-volume day does not necessarily have high turnover if cases are long and consecutive (few room resets needed).

The XGBoost models predict the **conditional mean** $\hat{\mu}_{p,d}$ of demand — the best single-number forecast given the 27 observed features. This is a regression problem optimized for minimum RMSE.

Why XGBoost for This Problem

- **Non-linear interactions.** The relationship between util_pct and future demand is non-linear: a provider at 90% util this week is likely to be at 90% next week (mean-reversion), but a provider at 90% for 12 consecutive weeks is different from one who hit 90% only last week. XGBoost captures these interactions through tree structure.
- **Missing data tolerance.** Some providers have fewer than 12 weeks of history for certain feature windows. XGBoost handles missing features gracefully without imputation.
- **Feature importance.** SHAP values from the XGBoost model feed Layer 3 explanations — every block change can be attributed to specific demand features.
- **Persistence via Optuna.** The SQLite study carries best hyperparameters across quarters. The model improves over time without starting from scratch each quarter.

Time-Series Cross-Validation Design

Standard k-fold CV cannot be used — it would leak future weeks into the training set. All splits must be temporal.

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║  TIME-SERIES CROSS-VALIDATION  ·  SITE-001  ·  H=13 weeks                     ║
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║                                                                               ║
║  EXPANDING WINDOW CV  (3 folds)                                               ║
║  ─────────────────────────────────────────────────────────────────────────    ║
║                                                                               ║
║  Fold 1:  Train W00–W07  │  Val W08–W09  │  gap: 0 weeks                      ║
║           ◄── 8 weeks ──►│◄── 2 weeks ──►│                                    ║
║                                                                               ║
║  Fold 2:  Train W00–W09  │  Val W10–W11  │  expanding: +2 train weeks         ║
║           ◄──── 10 wks ──►│◄── 2 wks ──►│                                     ║
║                                                                               ║
║  Fold 3:  Train W00–W11  │  Val W12     │  expanding: +2 train weeks          ║
║           ◄──── 12 wks ───►│◄── 1 wk ──►│                                     ║
║                                                                               ║
║  CV RMSE = mean(RMSE_fold1, RMSE_fold2, RMSE_fold3)  per provider             ║
║                                                                               ║
║  WHY EXPANDING WINDOW, NOT SLIDING WINDOW?                                    ║
```

```
║  With only H=13 weeks, a sliding window of fixed size would waste data.       ║
║  Expanding window uses all available training data for each fold —            ║
║  important when most providers have only 13 weeks of history.                 ║
║                                                                               ║
║  FINAL MODEL FIT:  Train on all W00–W12 with best hyperparameters.            ║
║                                                                               ║
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```

Hyperparameter Optimization via Optuna

Optuna TPE (Tree-structured Parzen Estimator) sampler with MedianPruner. The SQLite study persists across quarterly runs — Optuna warm-starts from the previous quarter's best params.

```
╔═══════════════════════════════════════════════════════════════════════════════╗
║  OPTUNA HYPERPARAMETER SEARCH  ·  CaseTime XGBoost model                      ║
╠═══════════════════════════════════════════════════════════════════════════════╣
║                                                                               ║
║  SEARCH SPACE                                                                 ║
║  ─────────────────────────────────────────────────────────────────────────    ║
║  Parameter             Range               Scale    Notes                     ║
║  ─────────────────────────────────────────────────────────────────────────    ║
║  n_estimators          [100, 1500]         int      Depth of ensemble         ║
║  max_depth             [3, 10]             int      Tree complexity           ║
║  learning_rate         [0.001, 0.30]       log      Shrinkage rate            ║
║  subsample             [0.50, 1.00]        float    Row sampling fraction     ║
║  colsample_bytree      [0.50, 1.00]        float    Column sampling           ║
║  min_child_weight      [1, 20]             int      Leaf size floor           ║
║  reg_alpha (L1)        [1e-8, 10.0]        log      Sparsity regularization   ║
║  reg_lambda (L2)       [1e-8, 10.0]        log      Ridge regularization      ║
║  gamma                 [0.0, 5.0]          float    Split gain threshold      ║
║                                                                               ║
║  PRUNER: MedianPruner                                                         ║
║  Prunes trials where intermediate RMSE exceeds median of completed            ║
║  trials at same step. Cuts ~40% of trials early — speeds search 2x.           ║
║                                                                               ║
║  BEST PARAMS FOUND (Q3 2026, SITE-001, CaseTime model):                       ║
║  ─────────────────────────────────────────────────────────────────────────    ║
║  n_estimators      =  420                                                     ║
║  max_depth         =  4                                                       ║
║  learning_rate     =  0.047                                                   ║
║  subsample         =  0.82                                                    ║
║  colsample_bytree  =  0.71                                                    ║
║  min_child_weight  =  8                                                       ║
║  reg_alpha         =  0.003                                                   ║
║  reg_lambda        =  1.24                                                    ║
║  gamma             =  0.21                                                    ║
```

```
║                                                                              ║
║  CV RMSE:  18.4 min  (CaseTime model, 3-fold average)                        ║
║  CV RMSE:   4.1 min  (Turnover model, 3-fold average)                        ║
║    Turnover is much more predictable than CaseTime — lower CV RMSE           ║
║    consistent with it being a function of case count (stable) rather         ║
║    than case complexity (variable).                                          ║
║                                                                              ║
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```

Point Forecast Results

```
╔══════════════════════════════════════════════════════════════════════════════╗
║  XGBOOST POINT FORECASTS  ·  SITE-001  ·  Selected providers  ·  Tuesday     ║
╠══════════════════════════════════════════════════════════════════════════════╣
║                                                                              ║
║  Provider         μ̂_casetime   μ̂_turnover   μ̂_total    Actual W12  Error     ║
║  ───────────────  ────────────  ────────────  ─────────  ──────────  ──────  ║
║  Dr. Garcia           81.6         13.2         94.8       79.1 min   -2.5   ║
║  Dr. Chen            432.0         48.0        480.0      441.0 min   +9.0   ║
║  Dr. Thompson         52.3          8.4         60.7       48.9 min   -3.4   ║
║  Dr. Kim             340.1         37.8        377.9      352.4 min  +12.3   ║
║  Dr. Wright            0.0          0.0          0.0        0.0 min    0.0   ║
║    (new hire — XGBoost has no history, predicts service-line mean)           ║
║  Dr. Torres          370.2         41.1        411.3      383.7 min  +13.5   ║
║  Dr. Patel           308.4         34.2        342.6      295.8 min  -12.6   ║
║                                                                              ║
║  FEATURE IMPORTANCE  (top 5, CaseTime model, SHAP mean |value|)              ║
║  ──────────────────────────────────────────────────────────────────────────  ║
║  Rank  Feature                    Mean |SHAP|  Interpretation                ║
║  ────  ─────────────────────────  ───────────  ──────────────────────────    ║
║   1    attn_weighted_util          28.4 min    Operating level dominates     ║
║   2    trailing_4w_mean_case       21.7 min    Recent history 2nd signal     ║
║   3    service_line_wk_mean        14.2 min    Peer context matters          ║
║   4    trailing_4w_std_case         9.1 min    Volatility reduces forecast   ║
║   5    exception_rate_series        6.8 min    Slot contestedness signal     ║
║                                                                              ║
║  OVERFITTING GUARD                                                           ║
║  ──────────────────────────────────────────────────────────────────────────  ║
║  Train RMSE (final model):  12.1 min                                         ║
║  CV RMSE (held-out folds):  18.4 min                                         ║
║  Generalization gap:  6.3 min  (acceptable — some overfitting to train)      ║
║  If gap > 15 min → increase min_child_weight, reduce max_depth               ║
║                                                                              ║
╚══════════════════════════════════════════════════════════════════════════════╝
```

Outputs

- `xgb_model_casetime.json`
- `xgb_model_turnover.json`
- `xgboost_best_params.json`
- `optuna_study.db` — persistent SQLite study across quarters
- $\hat{\mu}_{\text{casetime}}[p,d]$ and $\hat{\mu}_{\text{turnover}}[p,d]$— per provider × day predictions

---

Stage 4: Hierarchical Bayesian Variance Model

What the Model Is Doing and Why

After XGBoost predicts the conditional mean $\hat{\mu}_{p,d}$, we need to know: **how wrong is that prediction likely to be, and by how much?**

The Bayesian model answers this by fitting a distribution to the XGBoost residuals:

$r_{p,d,t} = \text{actual}_{p,d,t} - \hat{\mu}_{p,d}$

The learned distribution of residuals, combined with the XGBoost mean, produces the full predictive distribution for future demand. The 200 scenarios are draws from this distribution.

XGBoost learns the **signal** — the systematic predictable part of demand. The Bayesian model learns the **noise** — the residual unpredictability that remains after XGBoost has done its best. σ_{pd} is the standard deviation of XGBoost's prediction errors for provider p on day d.

Full Model Architecture

Three-Level Hierarchy

```
╔══════════════════════════════════════════════════════════════════════════════╗
║  HIERARCHICAL MODEL STRUCTURE  ·  Three levels of partial pooling            ║
╠══════════════════════════════════════════════════════════════════════════════╣
║                                                                              ║
║  SITE LEVEL  (one hyperparameter per site)                                   ║
║  ─────────────────────────────────────────────────                           ║
```

```
║  σ_site  ~  HalfCauchy(5.0)                                                   ║
║  Governs overall scale of residuals at SITE-001.                              ║
║  HalfCauchy: heavy tails — allows σ_site to be large if the site is           ║
║  genuinely unpredictable, without forcing it to be large.                     ║
║                                                                               ║
║  SERVICE LINE LEVEL  (one per service line)                                   ║
║  ─────────────────────────────────────────────────                            ║
║  σ_sl[sl]  ~  HalfNormal(σ_site)    for sl ∈ {Cardiac, Ortho, General}        ║
║  Each service line gets its own σ. Cardiac providers share one pool.          ║
║  Ortho providers share another. Cross-service pooling via σ_site.             ║
║                                                                               ║
║  PROVIDER × DAY LEVEL  (one per provider × day-of-week)                       ║
║  ─────────────────────────────────────────────────                            ║
║  σ_pd[p,d]  ~  HalfNormal(σ_sl[service_line(p)])                              ║
║  Each provider × day gets its own σ. Garcia's Tuesday ≠ Garcia's Monday.      ║
║                                                                               ║
║  SITE-001 PARAMETER COUNT                                                     ║
║  ─────────────────────────────────────────────────                            ║
║  σ_site                 :  1 scalar                                           ║
║  σ_sl × 3 service lines :  3 scalars                                          ║
║  σ_pd × 18 × 5 days     :  90 scalars                                         ║
║  Total latent variables  :  94                                                ║
║                                                                               ║
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```

Formal Model Statement

Site-level hyperprior:

$$\sigma_{\text{site}} \sim \text{HalfCauchy}(5.0)$$

Service-line-level prior:

$$\sigma_{\text{sl}}[\text{sl}] \sim \text{HalfNormal}(\sigma_{\text{site}}) \quad \forall\, \text{sl}$$

Provider×day-level prior:

$$\sigma_{pd}[p, d] \sim \text{HalfNormal}\big(\sigma_{\text{sl}}[\text{service\_line}(p)]\big) \quad \forall\, p,\ d$$

Likelihood (current — Normal on residuals):

$$r_{p,d,t} \sim \mathcal{N}(0,\ \sigma_{pd}[p, d]) \quad \forall\, p,\ d,\ t$$

Partial Pooling Effect

```
╔═══════════════════════════════════════════════════════════════════════════════╗
║  PARTIAL POOLING  ·  How data sparsity affects σ_pd posteriors                ║
╠═══════════════════════════════════════════════════════════════════════════════╣
║                                                                               ║
```

```
║  Provider       N residuals  σ_pd post. mean  Source of information        ║
║  ─────────────  ───────────  ───────────────  ─────────────────────────    ║
║  Dr. Chen           13       8.3 min           ~95% own data               ║
║  Dr. Garcia         13       25.5 min          ~95% own data (genuinely    ║
║                                                 high variance)             ║
║  Dr. Thompson        7       21.4 min          ~70% own / 30% Ortho prior  ║
║  Dr. Wright          0       18.9 min          100% Cardiac prior          ║
║                               (new hire — no residuals to fit)             ║
║  Dr. Kim            13       14.2 min          ~95% own data               ║
║                                                                            ║
║  SERVICE LINE POSTERIORS                                                   ║
║  σ_sl[Cardiac]  mean=17.1 min  (Garcia inflates Cardiac σ)                 ║
║  σ_sl[Ortho]    mean=15.8 min                                              ║
║  σ_sl[General]  mean=12.3 min  (most data, tightest)                       ║
║  σ_site         mean=14.9 min                                              ║
║                                                                            ║
║  KEY POINT: Wright's σ_pd = 18.9 min is principled, not arbitrary.         ║
║  It is the posterior expectation for a new Cardiac provider given all      ║
║  we know about Cardiac providers at this site. When Wright generates       ║
║  his first cases next quarter, his σ_pd updates toward his own data.       ║
║                                                                            ║
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```

Likelihood Family Selection

Why Normal Is a Convenient Lie

Normal residuals have three violations for OR demand:

Negative support: Demand cannot be negative. Normal allows $\hat{\mu} + \varepsilon < 0$ (clipped to zero). For Garcia ($\hat{\mu} \approx 1.4$ hrs/wk), this happens frequently, creating a zero spike the model doesn't acknowledge.

Symmetric tails: A provider can run far above mean on a busy week. They cannot run far below zero. Right skew is structural, not noise.

No structural zeros: Garcia has 23% zero-operation weeks. These are not "very low Normal draws" — they are a separate generating event (did he operate at all?).

Candidate Likelihood Families

Log-Normal:

$$\log(\text{actual}_{p,d,t} + \epsilon) \sim \mathcal{N}\left(\log(\hat{\mu}_{p,d}),\ \sigma_{pd}\right)$$

Multiplicative noise model. Strictly positive. Right-skewed. NUTS-friendly. Best default upgrade from Normal. Handle zeros with small ϵ floor or separate zero-inflation component.

Gamma:

$$\text{actual}_{p,d,t} \sim \text{Gamma}(\alpha_{pd},\ \beta_{pd})$$

with constraint $\alpha_{pd}/\beta_{pd} = \hat{\mu}_{p,d}$ *so the XGBoost mean is preserved. Shape* α_{pd} controls skew: large α → near-Normal (Chen); small α heavily right-skewed (Garcia when he operates). Strictly positive, no zero mass.

Weibull:

$$\text{actual}_{p,d,t} \sim \text{Weibull}(k_{pd},\ \lambda_{pd})$$

Shape k_{pd} directly controls hazard structure. $k < 1$: intermittent demand, many near-zero weeks (Garcia-type). $k > 1$: demand builds toward a reliable level (Chen-type). Most flexible single-family choice.

Zero-Inflated Gamma:

$$\text{actual}_{p,d,t} = \begin{cases} 0 & \text{with probability } \pi_{pd} \\ \text{Gamma}(\alpha_{pd}, \beta_{pd}) & \text{with probability } 1 - \pi_{pd} \end{cases}$$

$$\pi_{pd} \sim \text{Beta}(\pi_{\text{sl}} \cdot \kappa_{\text{sl}},\ (1 - \pi_{\text{sl}}) \cdot \kappa_{\text{sl}})$$

$$\pi_{\text{sl}} \sim \text{Beta}(1.0,\ 9.0)$$

Separates "did Garcia operate this week?" (Bernoulli with probability π_{pd}) from "given he operated, how much?" (Gamma). Correct generative model for providers with exception_rate > 0.15.

```
╔══════════════════════════════════════════════════════════════════════════════╗
║  LIKELIHOOD FAMILY SELECTION GUIDE  ·  Per-provider decision rules           ║
╠══════════════════════════════════════════════════════════════════════════════╣
║                                                                              ║
║  INPUTS per provider p, day d:                                               ║
║  ───────────────────────────────────────────────────                         ║
║  zero_week_fraction  =  fraction of weeks with actual ≈ 0                    ║
║  coeff_of_variation  =  std(actual) / mean(actual)                           ║
║  exception_rate      =  from Pre-Layer                                       ║
║                                                                              ║
║  DECISION TREE                                                               ║
║  ──────────────────────────────────────────────────────────────────────────  ║
║                                                                              ║
```

```
║  zero_week_fraction > 0.15?  ─── YES ───►  Zero-Inflated Gamma                ║
║         │                                                                      ║
║        NO                                                                      ║
║         │                                                                      ║
║         ▼                                                                      ║
║  coeff_of_variation > 0.50?  ─── YES ───►  Log-Normal or Weibull              ║
║         │                                                                      ║
║        NO                                                                      ║
║         │                                                                      ║
║         ▼                                                                      ║
║  coeff_of_variation > 0.20?  ─── YES ───►  Gamma                              ║
║         │                                                                      ║
║        NO                                                                      ║
║         ▼                                                                      ║
║                             Normal on residuals                                ║
║                                                                                ║
║  SITE-001 ASSIGNMENTS                                                          ║
║  ──────────────────────────────────────────────────────────────────────────    ║
║  Provider          zero_frac   CV     Family               Dev CRPS            ║
║  ───────────────  ──────────  ──────  ───────────────────  ──────────────      ║
║  Dr. Garcia        0.23       1.42   Zero-Inflated Gamma    4.1 min            ║
║  Dr. Thompson      0.08       0.71   Log-Normal             5.8 min            ║
║  Dr. Wright        N/A  (new)  —     ZI-Gamma (inherited)   N/A                ║
║  Dr. Chen          0.00       0.09   Normal                 1.2 min            ║
║  Dr. Kim           0.00       0.18   Normal                 2.1 min            ║
║  Dr. Torres        0.00       0.24   Gamma                  3.1 min            ║
║  Dr. Patel         0.03       0.38   Gamma                  4.4 min            ║
║  Dr. Jones         0.00       0.14   Normal                 1.8 min            ║
║  (10 others)       0.00–0.02  0.10–  Normal                 1.4–2.8 min        ║
║                               0.19                                             ║
║                                                                                ║
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```

Prior Specification

Principles: Priors should be weakly informative — not flat (causes pathological sampling) but not so tight they overwhelm data. Based on OR domain knowledge: residuals of 5 min are small, 20 min is moderate, 60 min is large.

$$\sigma_{\text{site}} \sim \text{HalfCauchy}(5.0) \quad (\text{50\% mass below 5 min, heavy tail to 60+ min})$$

$$\sigma_{\text{sl}} \sim \text{HalfNormal}(\sigma_{\text{site}}) \quad (\text{more concentrated at lower level})$$

$$\sigma_{pd} \sim \text{HalfNormal}(\sigma_{\text{sl}}) \quad (\text{tightest at leaf level})$$

For zero-inflation:

$$\pi_{\text{sl}} \sim \text{Beta}(1.0,\ 9.0) \quad (\text{mean=10\%, 90\% mass below 25\%})$$

$$\kappa_{\text{sl}} \sim \text{Gamma}(2.0,\ 0.1) \quad (\text{mean=20, moderate pooling strength})$$

Neal's Funnel and Non-Centered Parameterization

The Problem

The centered parameterization $\sigma_{pd} \sim \text{HalfNormal}(\sigma_{\text{sl}})$ creates a geometric pathology: when σ_{sl} is small, the valid range for σ_{pd} is tiny. NUTS must take extremely small steps in this narrow region, accumulating divergences and producing poorly-mixed chains.

```
╔═══════════════════════════════════════════════════════════════════════════════╗
║  NEAL'S FUNNEL  ·  Why centered parameterization fails                        ║
╠═══════════════════════════════════════════════════════════════════════════════╣
║                                                                               ║
║  POSTERIOR GEOMETRY when σ_sl is uncertain (centered parameterization):       ║
║                                                                               ║
║  σ_sl = 5.0   │  σ_pd can range [0, ~15]   ← WIDE, easy to sample             ║
║               │  ┌─────────────────────────────────┐                          ║
║               │  │                                 │                          ║
║               ▼  └─────────────────────────────────┘                          ║
║  σ_sl = 1.0   │  σ_pd can range [0, ~3]    ← NARROW, harder                   ║
║               │  ┌────────┐                                                   ║
║               ▼  └────────┘                                                   ║
║  σ_sl = 0.1   │  σ_pd can range [0, ~0.3]  ← FUNNEL NECK                      ║
║               │  ┌─┐  NUTS gets stuck here                                    ║
║               ▼  └─┘  gradients explode near σ_pd ≈ 0                         ║
║                                                                               ║
║  OBSERVABLE SYMPTOMS                                                          ║
║  Divergences   : > 50 / 1000 samples   (> 5%)                                 ║
║  ESS_bulk σ_pd : < 100  (out of 2000 draws)                                   ║
║  R-hat σ_pd    : > 1.10                                                       ║
║                                                                               ║
╚═══════════════════════════════════════════════════════════════════════════════╝
```

The Fix: Non-Centered Parameterization (NCP)

Separate the scale from the location of the prior by introducing standardized auxiliary variables:

Centered (do not use):

$$\sigma_{pd} \sim \text{HalfNormal}(\sigma_{\text{sl}})$$

Non-centered (use this):

$$\tilde{\sigma}_{pd} \sim \mathrm{HalfNormal}(1.0)$$

$$\sigma_{pd} = \sigma_{\mathrm{sl}} \cdot \tilde{\sigma}_{pd}$$

The sampler works on $\tilde{\sigma}_{pd}$ *with a fixed* $\mathrm{HalfNormal}(1.0)$ *prior independent of* σ_{sl}. The funnel disappears. Applied at all three levels:

$$\tilde{\sigma}_{\mathrm{sl}} \sim \mathrm{HalfNormal}(1.0), \quad \sigma_{\mathrm{sl}} = \sigma_{\mathrm{site}} \cdot \tilde{\sigma}_{\mathrm{sl}}$$

$$\tilde{\sigma}_{pd} \sim \mathrm{HalfNormal}(1.0), \quad \sigma_{pd} = \sigma_{\mathrm{sl}} \cdot \tilde{\sigma}_{pd}$$

```
╔══════════════════════════════════════════════════════════════════════════════╗
║  NCP IMPLEMENTATION  ·  NumPyro pseudocode                                   ║
╠══════════════════════════════════════════════════════════════════════════════╣
║                                                                              ║
║  CENTERED  (avoid)              NON-CENTERED  (use this)                     ║
║  ───────────────────────────    ─────────────────────────────────────        ║
║  sigma_site = sample(           sigma_site = sample(                         ║
║    "sigma_site",                  "sigma_site",                              ║
║    HalfCauchy(5.0))               HalfCauchy(5.0))                           ║
║                                                                              ║
║  sigma_sl = sample(             sigma_sl_raw = sample(                       ║
║    "sigma_sl",                    "sigma_sl_raw",                            ║
║    HalfNormal(sigma_site))        HalfNormal(1.0),                           ║
║                                   sample_shape=(n_sl,))                      ║
║                                 sigma_sl = sigma_site * sigma_sl_raw         ║
║                                                                              ║
║  sigma_pd = sample(             sigma_pd_raw = sample(                       ║
║    "sigma_pd",                    "sigma_pd_raw",                            ║
║    HalfNormal(sigma_sl))          HalfNormal(1.0),                           ║
║                                   sample_shape=(n_p, n_d))                   ║
║                                 sigma_pd = sigma_sl[sl_idx] * sigma_pd_raw   ║
║                                                                              ║
║  EXPECTED IMPROVEMENT                                                        ║
║  ────────────────────────────────────────────────────────────────────        ║
║  Metric          Centered    Non-Centered   Target                           ║
║  ──────────────  ──────────  ─────────────  ────────                         ║
║  Divergences     > 50/1000   < 5/1000       0                                ║
║  ESS_bulk σ_pd   < 100       > 400          > 400                            ║
║  ESS_tail σ_pd   < 50        > 200          > 200                            ║
║  R-hat σ_pd      > 1.10      < 1.05         < 1.01                           ║
║                                                                              ║
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```

Additional Sampling Strategies

Target acceptance probability: Set `target_accept_prob=0.90` (default 0.80). Higher target forces smaller step sizes, reducing residual divergences at the cost of slower mixing in easy regions. Worthwhile for hierarchical models.

Warmup steps: Increase to 1000 (default 500). With 94 latent variables, the mass matrix adapter needs more data to estimate the posterior geometry.

Four chains: Run 4 chains instead of 2 for more reliable R-hat estimates and better detection of multimodality.

MCMC Diagnostics

R-hat

Measures convergence by comparing within-chain variance to between-chain variance:

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}$$

Thresholds: $\hat{R} < 1.01$ excellent · $\hat{R} < 1.05$ acceptable · $\hat{R} \geq 1.05$ **do not use**.

ESS Bulk and ESS Tail

ESS_bulk governs accuracy of posterior mean and median. **ESS_tail** governs accuracy of P5 and P95 — the quantities we care about for chance constraints.

Thresholds: ESS_bulk > 400 per parameter · ESS_tail > 200 per parameter (from 4000 total draws).

Divergence Count

Each divergence indicates a posterior region the sampler failed to explore. Target: 0. Acceptable: < 10 out of 4000 draws. Unacceptable and must fix: > 10.

```
╔══════════════════════════════════════════════════════════════════════════════╗
║  MCMC DIAGNOSTICS REPORT  ·  SITE-001  ·  Q3 2026  ·  NCP  ·  4 chains       ║
╠══════════════════════════════════════════════════════════════════════════════╣
║                                                                              ║
║  CONFIGURATION                                                               ║
║  Sampler          : NUTS (NumPyro)                                           ║
║  Chains           : 4     Warmup: 1000 steps     Samples: 1000/chain         ║
║  Total draws      : 4000  Target accept: 0.90    NCP: YES                    ║
║                                                                              ║
║  GLOBAL SUMMARY                                                              ║
║  ──────────────────────────────────────────────────                          ║
║  Total divergences            :     0         ✓  (target: 0)                 ║
```

```
║  Max R-hat (all 94 params)    :  1.003        ✓  (target: < 1.05)               ║
║  Min ESS_bulk (all params)    :   587         ✓  (target: > 400)                ║
║  Min ESS_tail (all params)    :   264         ✓  (target: > 200)                ║
║                                                                                 ║
║  PER-PARAMETER DIAGNOSTICS  (selected)                                          ║
║  ─────────────────────────────────────────────────────────────────────────      ║
║  Parameter                     Mean   SD    R-hat  ESS_bulk  ESS_tail           ║
║  ───────────────────────────  ─────  ─────  ─────  ────────  ────────           ║
║  σ_site                        14.9   3.2   1.001    1,842     1,120            ║
║  σ_sl[Cardiac]                 17.1   4.1   1.002    1,503       892            ║
║  σ_sl[Ortho]                   15.8   3.8   1.001    1,612       934            ║
║  σ_sl[General]                 12.3   2.9   1.001    1,788     1,043            ║
║  σ_pd[Garcia, Tue]             25.5   5.2   1.003      847       412            ║
║  σ_pd[Garcia, Mon]             23.1   4.8   1.002      921       447            ║
║  σ_pd[Chen, Mon]                8.3   1.8   1.001    1,203       687            ║
║  σ_pd[Wright, Tue]             18.9   6.1   1.004      612       287  ◄min      ║
║    ↑ Wright lowest ESS — fewest data, most reliant on prior                     ║
║  σ_pd[Thompson, Mon]           21.4   5.8   1.003      734       341            ║
║  π_pd[Garcia, Tue]              0.21  0.08  1.002      934       481            ║
║    ↑ ZI component: Garcia Tuesday zero-operation probability = 21%              ║
║                                                                                 ║
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```

Train / Dev / Test Splitting

All splits are temporal — no random sampling.

$$\text{Train} = W_0, \ldots, W_7 \quad (\text{8 weeks, 60\%})$$

$$\text{Dev} = W_8, W_9 \quad (\text{2 weeks, 15\%})$$

$$\text{Test} = W_{10}, W_{11}, W_{12} \quad (\text{3 weeks, touched once for final evaluation})$$

Dev set uses: likelihood family selection (CRPS comparison across Normal/Log-Normal/Gamma/ZI-Gamma), prior hyperparameter tuning, coverage calibration check.

Test set uses: final held-out coverage report, CRPS per provider, alpha-consistency verification.

ECDF Calibration Check

For each actual observation $y_{p,d,t}$ in the dev set, compute its percentile rank within the posterior predictive:

$$z_{p,d,t} = F_{\text{pred}}\left(y_{p,d,t}\right) = \frac{1}{S} \sum_{s=1}^{S} \mathbf{1}\left[x_{p,d,t}^{(s)} \leq y_{p,d,t}\right]$$

If calibrated, z values are uniform on $[0,1]$ — ECDF is the diagonal. The KS statistic:

$$\mathrm{KS}_p = \sup_{z\in[0,1]}\left|\hat{F}_{z,p}(z) - z\right|$$

Flag providers with $\mathrm{KS}_p > 0.15$. Run both aggregate and per-provider.

```
╔═══════════════════════════════════════════════════════════════════════════════╗
║  ECDF CALIBRATION  ·  SITE-001  ·  Dev W08–W09  ·  Normal vs Per-family       ║
╠═══════════════════════════════════════════════════════════════════════════════╣
║                                                                               ║
║  NORMAL MODEL (all providers — uniform family)                                ║
║  ─────────────────────────────────────────────────                           ║
║  1.0 ┤                                         ┌──────────                    ║
║      │                                     ┌───┘                              ║
║  0.8 ┤                               ┌─────┘  S-shape deviation               ║
║      │                         ┌─────┘                                        ║
║  0.6 ┤                   ┌─────┘      · · · · · · IDEAL diagonal              ║
║      │             ┌─────┘      · ·                                           ║
║  0.4 ┤       ┌─────┘      · ·                                                 ║
║      │ ┌─────┘     · ·                                                        ║
║  0.2 ┤─┘    · ·                                                               ║
║  0.0 ┼──────────────────────────────────────────                              ║
║      0.0       0.5       1.0                                                  ║
║                                                                               ║
║  Above diagonal in lower half → too many actuals in predicted low tail        ║
║  Below diagonal in upper half → actuals overflow predicted high tail          ║
║  Cause: Normal underestimates right tail for Garcia, overestimates for        ║
║         Chen (Normal assumes symmetric errors both ways)                      ║
║                                                                               ║
║  PER-FAMILY MODEL (ZI-Gamma/Log-Normal/Gamma/Normal per provider)             ║
║  ─────────────────────────────────────────────────                           ║
║  1.0 ┤                                    ┌──────                             ║
║      │                                ┌───┘ ·  ·  IDEAL                       ║
║  0.8 ┤                           ┌────┘  ·  ·                                 ║
║      │                      ┌────┘  ·  ·                                      ║
║  0.6 ┤                 ┌────┘  ·  ·                                           ║
║      │            ┌────┘  ·  ·                                                ║
║  0.4 ┤       ┌────┘  ·  ·                                                     ║
║      │   ┌───┘  ·  ·                                                          ║
║  0.2 ┤ ┌─┘  ·  ·                                                              ║
║  0.0 ┼──────────────────────────────                                          ║
║      0.0       0.5       1.0                                                  ║
║                                                                               ║
║  Empirical ECDF closely tracks ideal diagonal ✓                               ║
║  KS max per provider: 0.11 (Thompson)   All < 0.15 threshold ✓                ║
║                                                                               ║
╚═══════════════════════════════════════════════════════════════════════════════╝
```

Full Validation Pipeline

```
╔══════════════════════════════════════════════════════════════════════════════╗
║  BAYESIAN VALIDATION PIPELINE  ·  Per-quarter execution                      ║
╠══════════════════════════════════════════════════════════════════════════════╣
║                                                                              ║
║  Phase 1 — PRIOR PREDICTIVE CHECK  (before data)                             ║
║  ─────────────────────────────────────────────────                           ║
║  □ Draw 1000 prior samples.                                                  ║
║  □ Implied demand range within [0, 480 min/week].                            ║
║  □ No draws produce physically impossible σ_pd.                              ║
║                                                                              ║
║  Phase 2 — LIKELIHOOD FAMILY SELECTION  (on dev set)                         ║
║  ─────────────────────────────────────────────────                           ║
║  □ Compute zero_week_fraction and CV per provider per day.                   ║
║  □ Assign candidate family using decision tree.                              ║
║  □ Fit each candidate on train set.                                          ║
║  □ Evaluate CRPS on dev set.                                                 ║
║  □ Select family with lowest average CRPS per provider.                      ║
║  □ Persist selection in config.json.                                         ║
║                                                                              ║
║  Phase 3 — PRIOR HYPERPARAMETER TUNING  (on dev set)                         ║
║  ─────────────────────────────────────────────────                           ║
║  □ Grid search: HalfCauchy scale ∈ {2, 5, 10},                               ║
║                 HalfNormal multiplier ∈ {0.5, 1.0, 2.0},                     ║
║                 Beta params ∈ {(1,4), (1,9), (1,19)}.                        ║
║  □ Select config minimizing CRPS on dev set.                                 ║
║  □ Persist in config.json.                                                   ║
║                                                                              ║
║  Phase 4 — FULL MODEL FIT  (on full train set)                               ║
║  ─────────────────────────────────────────────────                           ║
║  □ Fit XGBoost. Compute residuals. Fit Bayesian model.                       ║
║  □ NUTS: 4 chains × 1000 warmup × 1000 samples.                              ║
║  □ NCP applied at all three hierarchy levels.                                ║
║  □ target_accept_prob = 0.90.                                                ║
║                                                                              ║
║  Phase 5 — MCMC DIAGNOSTICS  (gate — must pass before proceeding)            ║
║  ─────────────────────────────────────────────────                           ║
║  □ Divergences ≤ 10 / 4000        FAIL → increase target_accept or warmup    ║
║  □ All R-hat < 1.05               FAIL → increase chains or warmup           ║
║  □ ESS_bulk > 400 per parameter   FAIL → increase sample count               ║
║  □ ESS_tail > 200 per parameter   FAIL → increase sample count               ║
║  If ANY check fails → do not generate scenarios.                             ║
║  Surface warning in calibration.json. Use fallback (α-scaling + 20% CV).     ║
║                                                                              ║
║  Phase 6 — COVERAGE CHECK  (on dev set)                                      ║
║  ─────────────────────────────────────────────────                           ║
║  □ Generate 200 posterior predictive scenarios for dev weeks.                ║
║  □ Coverage at 50/80/95% ≥ nominal level (conservative OK).                  ║
║  □ Per-provider KS test: max |ECDF(z) - z| < 0.15.                           ║
```

```
║                                                                          ║
║  Phase 7 — ALPHA-CONSISTENCY CHECK  (on dev set)                         ║
║  ─────────────────────────────────────────────────                       ║
║  □ drift_p = |E[demand_s] - α·hist_demand| / α·hist_demand               ║
║  □ PASS: drift_p < 0.15   FAIL: drift_p ≥ 0.15 → "drift_warning" flag    ║
║                                                                          ║
║  Phase 8 — FINAL TEST EVALUATION  (touch test set once)                  ║
║  ─────────────────────────────────────────────────                       ║
║  □ Final coverage, CRPS, KS on W10–W12.                                  ║
║  □ Compare to dev metrics. Flag if test-dev gap > 20% on CRPS.           ║
║  □ Record in calibration.json as "test_set_evaluation".                  ║
║                                                                          ║
╚══════════════════════════════════════════════════════════════════════════╝
```

Outputs

- `posterior.nc` or `posterior_idata.pkl` — full posterior samples
- `mcmc_samples.pkl` — {sigma_pd_casetime, sigma_pd_turnover} shape (draws, P, D)
- Convergence report in `calibration.json`

Stage 5: Scenario Generation (S = 200)

Two-Stage Sampling

For each scenario $s \in 1, \ldots, 200$:

Stage A — draw uncertainty level from posterior:

$$\hat{\sigma}^{(s)}_{pd,\text{case}} \sim p\left(\sigma^{\text{case}}_{pd} \mid \text{data}\right)$$

$$\hat{\sigma}^{(s)}_{pd,\text{turn}} \sim p\left(\sigma^{\text{turn}}_{pd} \mid \text{data}\right)$$

Stage B — draw demand realization:

$$\varepsilon_s \sim \mathcal{N}\left(0,\ \hat{\sigma}^{(s)}_{pd,\text{case}}\right)$$

$$\eta_s \sim \mathcal{N}\left(0,\ \hat{\sigma}^{(s)}_{pd,\text{turn}}\right)$$

$$\text{demand\_casetime}^{(s)}_{p,d} = \max\ \left(0,\ \hat{\mu}_{\text{casetime}}[p, d] + \varepsilon_s\right)$$

$$\text{demand\_turnover}^{(s)}_{p,d} = \max\ (0,\ \hat{\mu}_{\text{turnover}}[p,d] + \eta_s)$$

Stage A propagates uncertainty about σ_{pd} itself — we are not certain how uncertain a provider is, only that their σ_{pd} has a posterior distribution. This produces scenarios that are wider than simple $\mathcal{N}(\hat{\mu}, \sigma_{\text{mean}})$ because the extra width captures model uncertainty on top of demand uncertainty.

Quarterly scaling:

$$Q^{(s)}_{\text{casetime}}[p] = \sum_d \text{demand\_casetime}^{(s)}_{p,d} \cdot n_{\text{weeks}}$$
$$Q^{(s)}_{\text{turnover}}[p] = \sum_d \text{demand\_turnover}^{(s)}_{p,d} \cdot n_{\text{weeks}}$$

EarlyRelease is deterministic — no scenario subscript:

$$\text{EarlyRelease}_{\text{projected}}[p] = \alpha \cdot \sum_{b \in B_p^{\text{hist}}} \text{EarlyReleaseDur}_b$$

```
╔══════════════════════════════════════════════════════════════════════════════╗
║  SCENARIO GENERATION  ·  SITE-001  ·  Garcia × Tuesday  ·  S=200             ║
╠══════════════════════════════════════════════════════════════════════════════╣
║                                                                              ║
║  XGBoost inputs:                                                             ║
║  μ̂casetime = 81.6 min/wk    μ̂turnover = 13.2 min/wk                          ║
║                                                                              ║
║  Posterior:                                                                  ║
║  σ_pd_case ~ N(25.5, 5.2) clipped positive                                   ║
║  σ_pd_turn ~ N(4.8, 0.8) clipped positive                                    ║
║                                                                              ║
║  FIRST 10 SCENARIO DRAWS                                                     ║
║  ────────────────────────────────────────────────────────────────────────    ║
║   s  │ σ̂case │  ε_s   │ demand_case │ σ̂turn │  η_s  │ demand_turn            ║
║  ────┼────────┼────────┼─────────────┼────────┼───────┼─────────────         ║
║   1  │  23.4  │  +14.2 │  95.8 min   │   5.1  │  -3.2 │  10.0 min            ║
║   2  │  28.7  │  -31.1 │  50.5 min   │   4.6  │  +2.1 │  15.3 min            ║
║   3  │  22.1  │   +8.3 │  89.9 min   │   4.9  │  -1.8 │  11.4 min            ║
║   4  │  24.9  │  -44.2 │  37.4 min   │   5.2  │  +4.3 │  17.5 min            ║
║   5  │  29.3  │  +19.7 │ 101.3 min   │   4.4  │  -0.9 │  12.3 min            ║
║   6  │  21.8  │  -62.4 │   0.0 min   │   5.0  │  -5.8 │   7.4 min            ║
║         ↑ scenario 6: negative draw clipped to 0 (zero-demand week)          ║
║   7  │  26.1  │   +3.5 │  85.1 min   │   4.7  │  +1.4 │  14.6 min            ║
║   8  │  24.3  │  +28.9 │ 110.5 min   │   5.3  │  -2.6 │  10.6 min            ║
║   9  │  27.6  │  -18.3 │  63.3 min   │   4.5  │  +3.7 │  16.9 min            ║
║  10  │  23.8  │  +11.2 │  92.8 min   │   4.8  │  -4.1 │   9.1 min            ║
║                                                                              ║
║  FULL S=200 DISTRIBUTION  (weekly demand hrs)                                ║
║  ────────────────────────────────────────────────                            ║
║  Component       P10      P25      P50      P75      P90    Units            ║
║  ───────────── ──────── ──────── ──────── ──────── ────── ─────────          ║
```

```
║  CaseTime        0.78     1.09     1.36     1.64     1.87    hrs/wk          ║
║  Turnover        0.11     0.17     0.22     0.27     0.32    hrs/wk          ║
║  Total           0.96     1.31     1.55     1.83     2.11    hrs/wk          ║
║  Total × 13 wks  12.5     17.0     20.1     23.8     27.4    hrs/quarter     ║
║                                                                              ║
║  EarlyRelease_projected[Garcia] = 316 min = 5.27 hrs  (deterministic)        ║
║                                                                              ║
╚══════════════════════════════════════════════════════════════════════════════╝
```

Outputs

- `demand_scenarios.json` — per provider×day: point predictions, sigma posterior means, 200 (casetime_min, turnover_min) scenario pairs
- `early_release_projected.json` — per provider: deterministic scalar in minutes

Stage 6: Calibration

Coverage Check

For each nominal level $L \in 50\%, 80\%, 95\%$:

$$\text{empirical\_coverage}_L = \frac{|\{(p,d,t) \in \text{holdout} : \text{actual}_{p,d,t} \in \text{PI}_L^{(p,d)}\}|}{|\text{holdout}|}$$

Target: empirical coverage $\geq L$ (conservative bias is safe — scenarios wider than needed, not narrower).

Alpha-Consistency Check

For each provider p:

$$\text{drift}_p = \frac{\left|\mathbb{E}[\text{demand}^{(s)}[p]] - \alpha \cdot \text{historical\_demand}[p]\right|}{\text{historical\_demand}[p]}$$

Pass if $\text{drift}_p < 0.15$. Flag if ≥ 0.15 — signals that the Bayesian posterior has drifted far from the α-scaling baseline, indicating possible model misspecification.

```
╔══════════════════════════════════════════════════════════════════════════════╗
║  CALIBRATION REPORT  ·  SITE-001  ·  Q3 2026  ·  Per-provider families       ║
╠══════════════════════════════════════════════════════════════════════════════╣
║                                                                              ║
```

```
║  COVERAGE CHECK  (dev set W08–W09, 180 provider-day-week observations)      ║
║  ─────────────────────────────────────────────────────────────────────────  ║
║  Nominal  │ All providers │ Garcia (ZI-G) │ Chen (Normal) │ Status           ║
║  ─────────┼───────────────┼───────────────┼───────────────┼────────────────  ║
║  50%      │   53.2%       │   52.1%       │   51.8%       │   ✓              ║
║  80%      │   83.7%       │   82.4%       │   82.9%       │   ✓              ║
║  95%      │   96.1%       │   95.8%       │   96.3%       │   ✓              ║
║                                                                             ║
║  ALPHA-CONSISTENCY CHECK                                                    ║
║  ─────────────────────────────────────────────────────────────────────────  ║
║  Provider          E[demand_s]    α-scale baseline    drift     Status      ║
║  ────────────────  ────────────   ────────────────    ──────    ──────────  ║
║  Dr. Garcia         1.36 hrs/wk    1.34 hrs/wk         1.5%    ✓ PASS       ║
║  Dr. Thompson       0.51 hrs/wk    0.49 hrs/wk         4.1%    ✓ PASS       ║
║  Dr. Chen           7.20 hrs/wk    7.18 hrs/wk         0.3%    ✓ PASS       ║
║  Dr. Torres         6.18 hrs/wk    6.21 hrs/wk         0.5%    ✓ PASS       ║
║  All 18 providers                                    max=5.2%    ✓ ALL PASS  ║
║                                                                             ║
║  ECDF KS TEST                                                               ║
║  ─────────────────────────────────────────────────                          ║
║  Max KS per provider: 0.11 (Thompson)    All < 0.15 ✓                       ║
║                                                                             ║
║  VERDICT:  Model is well-calibrated. Proceed to scenario generation.        ║
║                                                                             ║
╚═════════════════════════════════════════════════════════════════════════════╝
```

Outputs

- `calibration.json` — coverage at 50/80/95%, CRPS per provider, KS statistics, alpha-consistency results, drift flags

---

Layer 1 Artifact Bundle

```
layer1_artifacts/run_YYYYMMDD_HHMMSS/
├── xgb_model_casetime.json          ← CaseTime XGBoost model
├── xgb_model_turnover.json          ← Turnover XGBoost model
├── xgboost_best_params.json         ← Best Optuna hyperparameters
├── optuna_study.db                  ← Persistent Optuna study (grows each quarter)
├── posterior.nc                     ← Full MCMC posterior (or .pkl fallback)
├── mcmc_samples.pkl                 ← {sigma_pd_casetime, sigma_pd_turnover}
│                                       shape: (draws=4000, P=18, D=5)
├── features.parquet                 ← 27-column feature matrix
├── residuals.parquet                ← XGBoost residuals per provider×day×week
├── indexers.json                    ← Provider and day index mappings
├── demand_scenarios.json            ← S=200 (casetime_min, turnover_min) per p×d
```

```
├── early_release_projected.json     ← Deterministic EarlyRelease per provider
├── attention_pooler.json            ← Saved w_q scalar
├── calibration.json                 ← Coverage, CRPS, KS, alpha-consistency
├── config.json                      ← All hyperparameters and family selections
└── SUMMARY.md                       ← Human-readable run summary
```

Next section: Layer 2 — Stochastic MIP + Pareto Frontier →

Document Owner: Zernitsky Itamar **Last Updated:** May 2026